



Phase 1 System Assessment: Water Treatment Plumbing, As-Marked

CLIENT	Happy Farm Botanicals Inc. (HFB Labs)	SERVICE PROVIDER	ClearWave Water Solutions
DOCUMENT NO.	CW-HFB-P1-2026-001, Rev A	ISSUE DATE	2026-09-21
RESPONSIBLE REPRESENTATIVE	Justin Shoemaker, Principal, ClearWave Water Solutions	STATUS	Draft for Client review
GOVERNING DOCUMENT	Quality Agreement between ClearWave Water Solutions and Happy Farm Botanicals Inc. (executed 2026)		

1. PURPOSE AND SCOPE

This document is the Phase 1 scoping deliverable for the HFB Labs water treatment plumbing redesign. Its job is limited and specific: **record what is installed today, as ClearWave marked it during the site walk.** It does not propose a new design. Redesign work is a separate phase and a separate work order.

It is issued as "Documentation" under Section 5 of the Quality Agreement and is built to satisfy the deliverable types named there: an assessment report, a line map, a component inventory, a functional status report, and a photo package.

Per Section 3 of the Quality Agreement, ClearWave does not perform water quality testing. Nothing in this document is a water quality result.

2. BASIS OF ASSESSMENT

Source	Status in this revision
ClearWave site walk with hand-marked line labels and flow arrows (38 photographs, files 1000034222 through 1000034263)	9 photographs reviewed and annotated in Rev A. Remaining 29 received; to be annotated in Rev B.
Facility plumbing drawing set, sheet series "03_PLUMBING" (22 sheets, supplied by Client)	Received. Not yet reconciled against field conditions. Reconciliation is a Rev B item.
ClearWave site walk video (Google Drive folder, supplied by ClearWave)	Received. Not yet reviewed against this document. Rev B item.
Quality Agreement, ClearWave / HFB Labs	Reviewed in full. Sections 2, 3, 5, 6, and 7 govern this deliverable.

Everything stated in Sections 3 through 6 below is a field observation from the photographs. Where a make, model, size, or setting could not be read from a photograph, it is written as "to be confirmed" rather than guessed.

3. SYSTEM OVERVIEW

The facility runs a single treatment train fed from the municipal supply. Reading the marked lines in flow order:

1. **City supply** enters at the line marked **CITY** / **MAIN IN**.
2. A **bypass manifold** in gray Schedule 80 PVC sits on the main, marked **MAIN IN** and **MAIN OUT**, with three red-handled true-union ball valves. This lets the treatment train be isolated while keeping supply to the building.
3. Water passes to a **carbon filter**, a fiberglass media tank marked **CARBON**, with an electronic control valve head. Its feed is marked **CARBON IN**.
4. Treated water feeds a **commercial reverse osmosis (RO) skid**. The RO feed line is marked **RO IN** and carries an actuated ball valve tied to an AquaGate leak controller. RO product leaves on the line marked **RO OUT**.
5. Downstream distribution runs overhead on insulated lines marked **DI**, **CITY OUT**, and **Recir** (recirculation return).

Whether a softener or a deionization (DI) polisher sits between the RO and the **DI** line could not be confirmed from the photographs reviewed in this revision. The **DI** label suggests a DI stage exists; its location and type are a Rev B confirmation item.

4. AS-MARKED LINE MAP

Each row is one line as ClearWave marked it in the field. "Material" is what is visible in the photograph. Pipe sizes are not recorded in this revision; they are a field-measurement item.

Line label (as marked)	Flow direction marked	Material observed	Insulation	Where observed	Photo ref
CITY	Arrow marked, into building	PVC with red-handle ball valve at branch	Foam, tape-wrapped	Mechanical wall, near flex conduit	1000034243
MAIN IN	Arrow marked	Cream CPVC, then gray Sch 80 PVC at manifold	None on CPVC	Bypass manifold and carbon feed	1000034224, 1000034253
MAIN OUT	Arrow marked	Gray Sch 80 PVC	None	Bypass manifold	1000034224
CARBON IN	Arrow marked	Gray Sch 80 PVC	None	Carbon tank inlet	1000034253
RO IN	Arrow marked, toward RO	Copper, insulated; actuated ball valve inline	Foam	Wall above AquaGate controller	1000034244
RO OUT	Arrow marked, away from RO	Copper, insulated; push-fit elbow at valve	Foam	Wall run with brass fitting and small clear tube	1000034245
DI	Arrow marked	Copper, insulated; sweat coupling visible	Foam	Wall run near solenoid valve; also overhead	1000034246, 1000034238, 1000034237
CITY OUT	Arrow marked	Insulated line, material not visible	Foam, tape-wrapped joints	Overhead along structure	1000034238
Recir	Arrow marked	Insulated line, material not visible	Foam, tape-wrapped joints	Overhead along structure	1000034238

4.1 FLOW DIAGRAM, AS MARKED

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CITY / MAIN IN
|
+---- [Sch 80 PVC bypass manifold: MAIN IN / MAIN OUT, 3 ball valves] ----> MAIN OUT (to building)
|
CARBON IN
|
[CARBON filter tank, electronic head]
|
RO IN ----[actuated ball valve, AquaGate leak controller]
|
[Commercial RO skid: prefilter, pump, 3 flow meters, 3 gauges]
|
RO OUT
|
DI (stage location to be confirmed)
|
overhead distribution: DI / CITY OUT / Recir (return)

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This is a text sketch of the marked lines only. It is not an engineered P&ID.

5. COMPONENT INVENTORY

#	Component	Marking / identifiers observed	Location	Condition noted	Photo ref
C-1	Bypass manifold	Gray Sch 80 PVC tee-and-loop; three red-handled true-union ball valves; unions on both legs	Main line, mechanical wall	Serviceable. Cardboard stored against piping.	1000034224
C-2	Carbon filter	Fiberglass media tank, hand-marked CARBON ; electronic control valve head with LCD and push buttons. Make, model, tank size, media volume: to be confirmed.	Mechanical wall	Serviceable. Power cord coiled and hanging at head.	1000034253
C-3	RO skid	Black-panel commercial RO unit: three rotameter flow meters, three pressure gauges, digital controller with LED status and a laminated quick-reference card, one blue 20 in. prefilter housing, stainless pump, two sample or drain valves. Make, model, rated GPD: to be confirmed.	Floor, on wood pallet	Operating position unclear. Open buckets staged under skid.	1000034222
C-4	RO feed isolation	Actuated ball valve, orange handle, wired to controller	RO IN line	Serviceable	1000034244
C-5	Leak controller	AquaGate wall-mount controller with quick-reference label; cable to C-4	Wall below RO IN	Serviceable	1000034244
C-6	Solenoid or actuated valve	White-body valve with wire lead, on line below DI	Wall below DI run	Function to be confirmed	1000034246
C-7	Push-fit fitting	Push-fit elbow on copper, at brass valve	RO OUT line	Serviceable. See observation O-4.	1000034245
C-8	Brass valve with small tube	Brass valve body with small clear tubing tee'd off	RO OUT line	Purpose to be confirmed (sample port or drain)	1000034245

#	Component	Marking / identifiers observed	Location	Condition noted	Photo ref
C-9	Pipe supports	Copper two-hole straps, single-hole clamps, threaded rod hanger	Wall runs and overhead	Serviceable	1000034245, 1000034246
C-10	Overhead distribution	Three parallel insulated lines along structural steel, marked DI , CITY OUT , Recir	Ceiling structure	Insulation joints tape-wrapped. See O-2.	1000034238
C-11	Large insulated service lines	Two large-diameter lines, black insulation with yellow-jacketed sections, overhead	Production area ceiling	Not part of treatment train; identity to be confirmed from 03_PLUMBING sheets.	1000034237

6. FUNCTIONAL STATUS AND OBSERVATIONS

Section 2 of the Quality Agreement requires ClearWave to notify the Client of any observed condition that may affect water quality, system safety, or regulatory compliance, even when outside the active scope. The observations below are made under that clause. They are field observations, not test results.

#	Observation	Why it matters	Suggested handling
O-1	All line identification is hand-written marker on foam insulation, with hand-drawn arrows.	Not compliant with ASME A13.1 (pipe identification), which Section 2 of the Quality Agreement names as the applicable standard. Marker on foam fades and is lost when insulation is replaced.	Formal labeling is proposed as a Phase 2 line item. See Section 7.
O-2	Foam insulation is split, gapped, or held with black tape at several joints and valve bodies.	Condensation on cold RO/DI lines can drip onto equipment and floors. Gaps also hide fittings from inspection.	Note for redesign scope. Not urgent.
O-3	Piping mixes four materials in one train: copper, cream CPVC, gray Sch 80 PVC, and push-fit fittings.	Mixed materials are workable, but each transition is a potential leak point and each needs the right fitting type. Copper downstream of RO is also a possible source of copper pickup in low-TDS water.	Confirm material of each segment in Rev B. Redesign should standardize.
O-4	Push-fit elbow used on the RO OUT line.	Push-fit fittings are acceptable on potable copper but are not typically specified for process water in a regulated facility.	Confirm with Client whether push-fit is acceptable under their quality system.
O-5	RO skid sits on a wood pallet with open buckets underneath.	Suggests either temporary placement or an unresolved drip or drain. Wood pallets in a wet area are a housekeeping and pest concern.	Confirm skid mounting and drain routing on next site walk.
O-6	Sample and drain lines near the RO are small clear tubing, unlabeled.	Unlabeled tubing can be mistaken for a sample point.	Include in labeling plan.
O-7	Location and type of the DI stage feeding the DI line could not be confirmed from the photographs reviewed.	The line map is incomplete without it.	Rev B item.

No unsafe condition requiring immediate work stoppage was observed in the photographs reviewed.

7. PIPE IDENTIFICATION PLAN (PROPOSED FOR PHASE 2)

Included here so the Client can see what "ASME A13.1 compliant" will look like. This is a proposal for scope, not work performed.

Line (as marked today)	Proposed legend	Color field (ASME A13.1)	Notes
CITY / MAIN IN	CITY WATER	Green field, white text	Potable water
MAIN OUT	CITY WATER TO BUILDING	Green field, white text	
CARBON IN	CARBON FILTER FEED	Green field, white text	
RO IN	RO FEED	Green field, white text	
RO OUT	RO PERMEATE	Green field, white text	Legend distinguishes it from feed
DI	DI WATER	Green field, white text	Confirm DI stage first (O-7)
Recir	DI RECIRC RETURN	Green field, white text	
RO reject / drain lines	RO REJECT TO DRAIN	Green field, white text	Not marked today; add
Sample tubing	SAMPLE POINT (numbered)	Tag, not pipe label	Number each point

Label size and placement follow ASME A13.1 Table 1 (letter height and label length by outside diameter) with labels at every valve, every wall or floor penetration, every change of direction, and at intervals along straight runs. Flow arrows on every label.

8. IQ / OQ FRAMEWORK OUTLINE

Section 5 of the Quality Agreement lists IQ and OQ framework documents as deliverables. This outline shows what each would verify for this system, so the Client can decide whether to include them in Phase 2.

Installation Qualification (IQ) would verify:

- Each component in Section 5 is installed per its manufacturer data sheet (make, model, serial recorded).
- Piping material, size, and joint type recorded for every segment on the line map.
- Every line labeled per Section 7 and every valve tagged.
- Drawings (03_PLUMBING and the ClearWave line map) match the installed condition, with a redline of every difference.
- Utilities (power to RO skid, carbon head, controllers, drain connections) confirmed.

Operational Qualification (OQ) would verify:

- Bypass manifold isolates the train and maintains building supply, valve by valve.
- Carbon head runs a full backwash and rinse cycle on its programmed schedule.
- RO skid reaches and holds its rated feed pressure and product flow; reject ratio recorded.
- AquaGate controller closes the RO feed valve on a simulated leak.
- Recirculation loop maintains flow with the DI stage in service.
- Alarm and shutdown functions on each controller confirmed.

Water quality acceptance criteria within OQ would be set and tested by the Client's own licensed personnel, per Section 3 of the Quality Agreement.

9. CHANGE CONTROL AND DEVIATIONS

- This document records existing conditions. It does not authorize any change to scope, materials, equipment, or procedures. Any change requires a written change order approved by both parties (Quality Agreement Section 6).
- If hidden plumbing, restricted access, or an unsafe condition is found during later work, ClearWave will pause that work, document the cause, and submit a written change order before continuing (Section 6).
- Any deviation identified during Services will be reported in writing to the Client's designated point of contact within two business days (Section 7).

10. PHOTO PACKAGE INDEX

All 38 photographs are retained by ClearWave under document CW-HFB-P1-2026-001 for a minimum of seven years (Section 5). The 9 annotated in this revision are listed with descriptions. The remaining 29 are listed by file number and will be annotated in Rev B.

File	Subject	Rev A annotation
1000034222	RO skid front panel: flow meters, gauges, controller, prefilter housing	Yes
1000034224	Sch 80 PVC bypass manifold, MAIN IN / MAIN OUT	Yes
1000034237	Production area overhead: large service lines, DI line	Yes
1000034238	Overhead run: DI , CITY OUT , Recir	Yes
1000034243	CITY branch with ball valve, taped insulation	Yes
1000034244	RO IN , actuated valve, AquaGate controller	Yes
1000034245	RO OUT , push-fit elbow, brass valve with tube	Yes
1000034246	DI line, sweat coupling, solenoid valve below	Yes
1000034253	Carbon tank and head, CARBON IN , MAIN IN CPVC	Yes
1000034223, 1000034225, 1000034226, 1000034228 through 1000034231, 1000034236, 1000034239 through 1000034242, 1000034247 through 1000034252, 1000034254 through 1000034261, 1000034263	Received	Rev B

11. OPEN ITEMS FOR REV B

#	Item	Owner
1	Annotate the remaining 29 photographs and update Sections 4 and 5	ClearWave

#	Item	Owner
2	Reconcile the 03_PLUMBING drawing set (22 sheets) against the field line map; issue a redline list	ClearWave
3	Review site walk video and add any lines or components not captured in photographs	ClearWave
4	Confirm make, model, serial, and size for C-2 and C-3; confirm function of C-6 and C-8	ClearWave, on site
5	Confirm location and type of the DI stage (O-7)	ClearWave, on site
6	Measure and record pipe sizes for every line in Section 4	ClearWave, on site
7	Confirm the identity of the two large overhead lines (C-11) from drawings	ClearWave
8	Confirm whether push-fit fittings are acceptable under the Client's quality system (O-4)	Client
9	Name the Client's designated point of contact for this project (Section 3)	Client
10	Decide whether Phase 2 includes formal labeling (Section 7) and IQ/OQ (Section 8)	Client

12. EXCLUSIONS

- No water quality testing, sampling, or analysis (Quality Agreement Section 3).
- No engineering design, sizing, or specification of replacement equipment or piping.
- No assessment of the building's general plumbing, drainage, or the large overhead service lines (C-11) beyond identifying them.
- No verification of code compliance of existing work performed by others.
- No opinion on regulatory status of the facility.

Sources

1. ClearWave site walk photographs, files 1000034222 through 1000034263, taken at the HFB Labs facility. Nine reviewed for this revision.
2. Quality Agreement, ClearWave Water Solutions and Happy Farm Botanicals Inc., executed 2026. Sections 2, 3, 5, 6, 7 cited.
3. ASME A13.1, Scheme for the Identification of Piping Systems, current edition, for Section 7.
4. Facility drawing set "03_PLUMBING," 22 sheets, supplied by Client. Not yet reconciled.

ClearWave Water Solutions · support@clearwavewater.com · 757-828-8594 · clearwavewater.com